

(Un)Common™ – HMT

HUMIDITY • MOISTURE • TEMPERATURE

MAKE YOUR OWN GRAIN BIN MONITORING SYSTEM

Project Ideas and Programming Code by

The Computer Connectors LLC



WHY SPEND THOUSANDS WHEN YOU COULD SPEND JUST A FEW HUNDRED?!

Now Supporting 24 Grain Types!

- Know the condition of your grain as you place it into your bin
 - HMT – Humidity, Moisture, Temperature
- Know the condition of your grain for the duration of its storage in your bin
- Minimize spoilage
- Avoid losing money by over drying your grain
- Make more money by (legally) re-hydrating your grain to (near) industry standards
- (Hopefully) Minimize the need to enter your bin to remove/breakup spoilage
- (And, did we mention)... **Make More Money**

Building the Device

Getting Started

If you are reading this, you are probably a farmer.

Be proud of that!

Nobody feeds and fuels the world like you.

There are a lot of people who are secretly jealous of you. Yes - Seriously!

Why else would they spend so much time and effort trying to tear us down?

Ok, We know...

Nobody likes to read instruction manuals. We feel the same way.

In this case, however, we strongly recommend reading this manual in its entirety – at least once prior to beginning the build process.

Reading this manual will help you get a good feel for:

- Where you are starting from
- A realistic goal
- The tools you will need
- The project parts you will need to order
- The building process
- And finally, what to expect at the end of the build.

Table of Contents

Contents

Getting Started.....	3
Table of Contents	4
Building the Device	6
A Word of Encouragement	6
Our Objective	6
A Request.....	6
Your Goal.....	7
A Few Pictures.....	7
Time Necessary to Build the Device.....	8
Parts You Will Need	9
Arduino MEGA 2560 Rev3 (with USB cable)	9
MEGA 2560 Enclosure Case (NO LONGER RECOMMENDED)	10
“Prototype Screw Terminal Block Shield”	10
Arduino MEGA compatible 2.8” TFT LCD touch screen	10
AM2305B Temperature and Humidity Sensors	11
4.7K Ohm Resistors	12
PRO Version Enablement Components [Required for PRO Version]	12
LEDs and 330 Ohm Resistors [PRO Version (Optional)]	12
Small box that you can use to protect and mount the Arduino MEGA and LCD	13
Electrical Junction Box (PVC).....	13
Electrical Outlet Box.....	14
Lengths of landscape or other type of low-voltage wire (length depends upon bin size)	14
Wire Splice Connectors or Terminal Block.....	14
1” PVC pipe (length depends upon bin size).....	15
Stainless steel cable to hang the sensor end of the PVC pipe inside the bin.	15
You will also need the necessary hardware to hang the PVC either along the sidewall or at the top of the bin and along the outside of the bin going down to the HMT device’s mounting location.....	16
Tools You Will Need	17
Assembling the Device	18

Schematic.....	18
Assembling the Pieces and Placing Them in the Box	19
Sensor Placement Decisions	21
Assembling and Mounting the Interior PVC Pipes	22
Assembling the Exterior PVC Pipes	26
Connecting the Sensors to the Low-Voltage Cable & PVC Pipe Assembly	27
Preparing the Sensors	27
Preparing the Sensors (Alternative Method).....	28
Measuring, Cutting, and Installing the Cable.....	29
Connecting the HMT Device to the 12 Position Terminal Block	31
Connecting the Sensors to the 12 Position Terminal Block	33
Connecting the Weather Sensor to the Terminal Block	33
Connecting Sensors 1 – 3 to the 12 Position Terminal Block.....	33
Connecting LED and Accessory Cable to Terminal Block	33
Admiring What You Have Accomplished	35
Installing the Firmware	37
Installing the “Pro” Version License (optional)	39
Preliminary Testing	39
Finalizing the Installation	40
Final Testing	40
Appendix A: Our Build and Installation Pictures	42
The HMT Device	42
Weather Sensor – Protected by Electrical Box	43
Hanging the Sensor PVC Pipes	44
Protecting Some of the Outside Equipment	46
A Word From Our Sponsor.....	47
Our Other Products and Services.....	47

Building the Device

A Word of Encouragement

We know – this document looks daunting. You might as well go climb Everest. 😊

However, the reason it looks so difficult is because we have tried to think of and include every bit of detail to help you prepare and complete the build.

Trust us – you can do it!

You may even find it enjoyable.

Our Objective

Our goal is to provide you with the information you need to successfully build your own grain monitoring system. On the pages that follow, we will work you through the process we used to build our systems, including: Some of the lessons we learned along the way, the parts we used, the assembly process, loading the firmware, and the bin installation.

Please note: Below are the instructions to build a PRO version of our HMT device. Realistically, we know that you will probably build the FREE version first as your own “proof of concept.” But, **if you build the system to accommodate the PRO version now**, you may purchase a PRO license and upgrade the FREE version at any time.

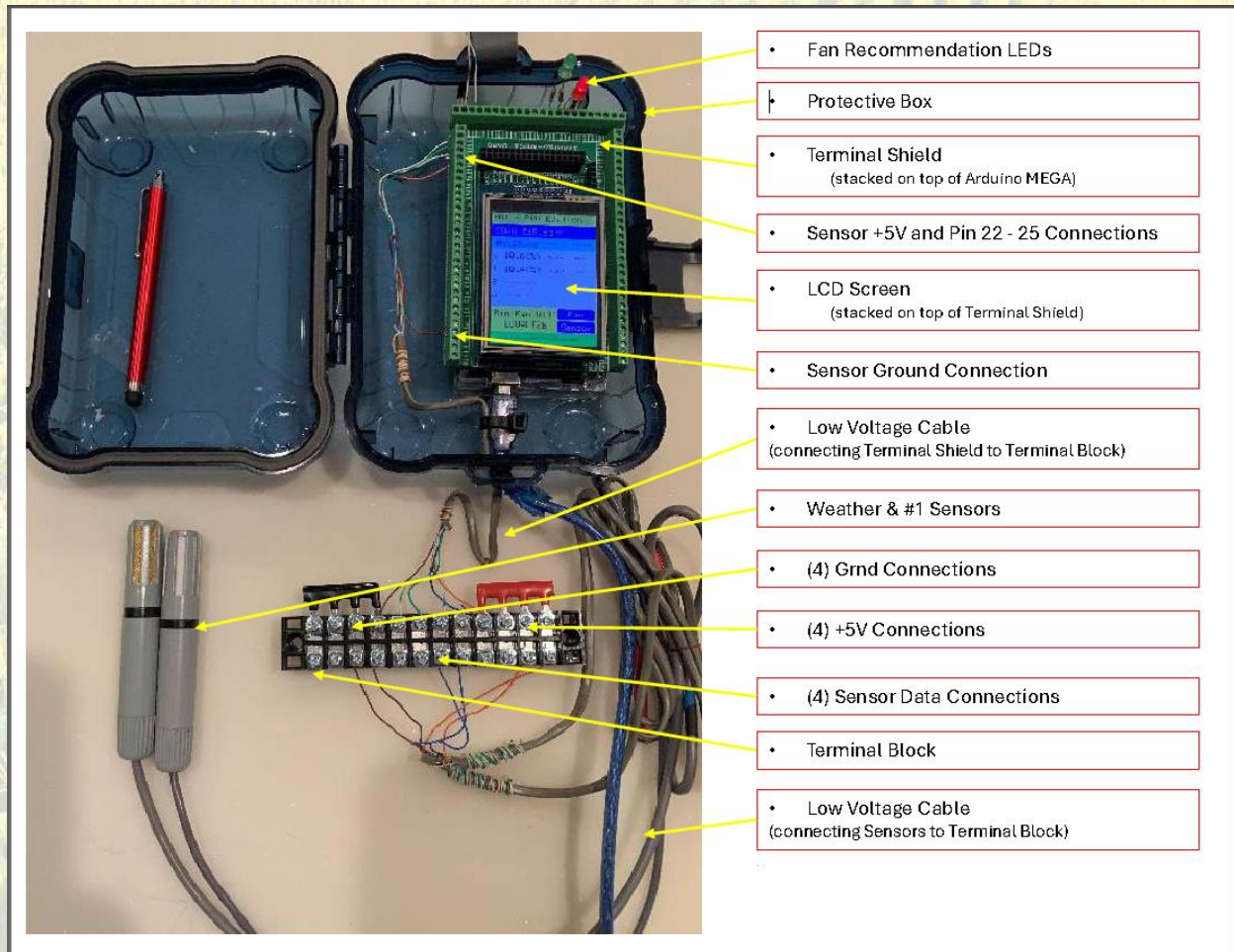
A Request

As you evaluate the instructions and your particular grain bin setups, you will undoubtedly think of better ways to complete the build and/or installation. Please feel free to share those ideas. We may even incorporate those ideas in future documentation.

Your Goal

A Few Pictures

If you are like us, you like pictures. One picture can make a whole lot more sense than even a few thousand words. Thus, we are including the following picture so that you may gain a mental image of the end goal.



Please visit Appendix C near the end of this document for more build and installation pictures.

Time Necessary to Build the Device

We would expect that, after gathering your parts, you could build and test the functionality of this device in less than one or two days. (Makes a great rainy/snowy day project!)

For the most part, building the device is minimally difficult. The most difficult part may be hanging the PVC pipe and sensors inside your bin. However, the device does contain some very small components.

Thus, please be patient and use a gentle hand!

Parts You Will Need

- Arduino MEGA 2560 Rev3
- “Prototype Screw Terminal Block Shield”
- Arduino MEGA compatible 2.8” TFT LCD touch screen
- AM2305B Temperature and Humidity Sensors
- LEDs and Resistors [PRO Version (optional)]
 - (1) Green LED and (1) Red LED
 - (2) 330 Ohm Resistors
- PRO Version Enablement Components [PRO Version (Required)]
- Small box that you can use to protect and mount the Arduino, LCD, and one AM2305B sensor
- Electrical Box
- Lengths of landscape, CAT5, or other type of low-voltage wire
- Wire Splice Connectors or Terminal Block
- 1” PVC pipe
- Stainless steel cable
- Hardware to hang the PVC pipe
- Junction box
- 12 position terminal block

Arduino MEGA 2560 Rev3 (with USB cable)



- Arduino.cc online store
- SKU A000067 / Barcode 7630049200067
 - <https://store-usa.arduino.cc/products/arduino-mega-2560-rev3?queryID=c6c29eb83a9ad4d733b9ffa8a0f1b326&selectedStore=us>
- Please note: There are “knock-off” varieties of this micro-computer. Some may work. Some may not. Your best luck is going to be with the official version (linked above).

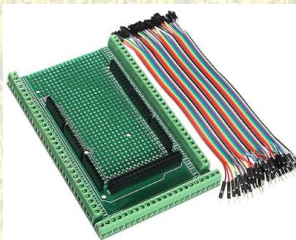
MEGA 2560 Enclosure Case (NO LONGER RECOMMENDED)



After a lot of unnecessary troubleshooting & frustration, we are no longer recommending this product.

We have had several instances where this case prevented solid electrical contact between the MEGA and the Terminal Shield. If you are using one of these cases on a previously build device and you have intermittent sensor or display issues, remove this case.

“Prototype Screw Terminal Block Shield”



- This will make your data wiring job WAY easier and more reliable.
- Amazon.com: Search for “ALAMSCN MEGA2560 PCB Prototype Screw Terminal Block Shield Board for Arduino MEGA 2560 Expansion with 40P Connect Wires”.

Arduino MEGA compatible 2.8” TFT LCD touch screen



NOTE: You may also find it helpful to have a “Touch Screen Pen.” Usually, one comes with the LCD screen, but they are kinda cheap and hard to use. Look around the house. You may even have a few nicer ones lying around.



- ~~Adafruit online store (Product ID: 376)~~
 - ~~<https://www.adafruit.com/product/376>~~
- Amazon Link
 - https://www.amazon.com/dp/B0BFDXTN9Z/?coliid=I2NAJ8I73EUKII&colid=3L6NS0IG4D4OF&psc=1&ref=lst_c_wl_lv_ov_lig_dp_it

IMPORTANT NOTE: It has recently come to our attention that Adafruit is selling an updated version of this touch screen. We have not tested that newer version. For now, if Adafruit is out of stock, please find and purchase the older version from another source (ex. Amazon). You will notice the difference by one of the description items.

Look for “8 bit digital interface, plus 4 control lines”.

DO NOT ORDER the newer version identified by “*The display uses digital pins 9, 10 and SPI. Touchscreen controller requires I2C and pin #2 optional IRQ. microSD pin requires digital #4*” in the description.

Even though Adafruit has stopped manufacturing the older version, we have been able to buy them from Amazon – see above link.

If that link doesn't work, you should be able to just search Amazon for some variation of the following product title: **“2.8” TFT LCD Display Module 5V/3.3V ILI - 9341 Resistive Touchscreen 320x240 8-bit Parallel Interface w/SD Card Slot for Arduino Mega-2560”**.

In the next iteration of the HMT device (probably 2026), we may swap the original LCD with a different (and larger) LCD.

That should be an easy upgrade...

- 1) Take old one off
- 2) Put new one on
- 3) Update the firmware.

AM2305B Temperature and Humidity Sensors



- (Quantity 2) When building the FREE version
- (Quantity 3 or 4) When building the PRO version

Note: Sometimes, these sensors come without any documentation. But, all you really need is the wire color code. All of the sensors we have ordered came with the following cable colors:

Red: +5V

White (or Black): Ground

Yellow (or Blue): Data

4.7K Ohm Resistors

For each sensor, you will need one of these resistors.



You can easily find these online.

PRO Version Enablement Components [Required for PRO Version]

If you chose to purchase a license to upgrade your device to the PRO version, simply order the upgrade from our website: www.TheComputerConnectors.com . We will then send you the necessary component(s) and the instructions for installing them.



LEDs and 330 Ohm Resistors [PRO Version (Optional)]



(330 Ohm Resistor quantity: 2)

As mentioned in the “PRO Features” section, the PRO version of this device has the ability to power four “outputs.” Two of those outputs are designed for LEDs (a.k.a. Diodes).

Make Your Own Grain Bin Monitoring System

Just for fun: During our research and development, we simply found some old electronics that we were going to recycle and that had the LEDs and resistors we want above. We then desoldered the LEDs and resistors from those old circuits and used them for the devices we used in our bins. You may do the same.

You may purchase those components from your favorite electronics store. Or, you may order those components from us during the PRO version order checkout process.

Small box that you can use to protect and mount the Arduino MEGA and LCD



- We used a simple box from Walmart, but you can use any box you like as long as it protects the components from rain and direct sunlight and it allows you to view the LCD screen without opening the box.
 - Additionally, always mount the device in a location that does NOT receive direct sunlight.
- <https://www.walmart.com/ip/Outdoor-Products-Small-Watertight-Dry-Box-Blue-Polycarbonate/50109790?athbdg=L1600&from=/search>
 - NOTE: There is a larger (deeper) version of this box if you prefer to have more room.
 - <https://www.walmart.com/ip/Outdoor-Products-Large-Watertight-Case-Dry-Box-Orange-8-x-6-75-x-3-5-Polycarbonate/36547548?fulfillmentIntent=Shipping&athbdg=L1600>

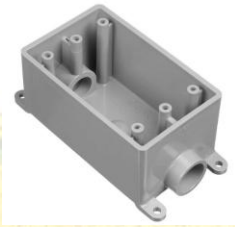
Electrical Junction Box (PVC)



- This will be used to house the connections between the HMT device and the sensors.

Make Your Own Grain Bin Monitoring System

Electrical Outlet Box



- This will be used to protect the “Weather” sensor.
- More on this later.

Lengths of landscape or other type of low-voltage wire (length depends upon bin size)

- You have options here!
- We typically use “Outdoor” rated CAT3, CAT5e, or CAT6 cable because it is both super easy to work with, and it is cheap.

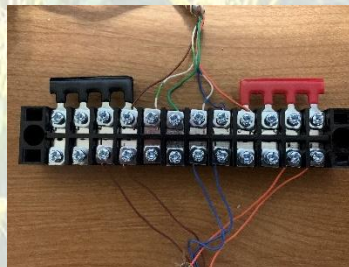


- Outdoor landscape cable may also be an option as long as it has 4 or more conductors.
- You could also use something like 4 conductor thermostat wire. HOWEVER, thermostat wire is typically NOT outdoor/UV resistant. Thus, it will not last very long outdoors.
- The quantity you need will be at least 10% to 20% more than the length of PVC you calculate (below).

Wire Splice Connectors or Terminal Block



or



- https://www.amazon.com/gp/product/B07NPGV7G2/ref=ppx_yo_dt_b_asin_title_o00_s00?ie=UTF8&psc=1
- https://www.amazon.com/gp/product/B07CLW5FPS/ref=ppx_yo_dt_b_asin_title_o01_s00?ie=UTF8&psc=1

Make Your Own Grain Bin Monitoring System

1" PVC pipe (length depends upon bin size)

- We will be using this to protect our wiring both inside and outside the bin.
- You basically have two options here.
- Basic Schedule 40 PVC or PVC Conduit.
 - The conduit type will probably last longer, but even in full-sun outside the bin, the Schedule 40 should be just fine.



- The amount you need will depend upon your individual installation. To get an estimate, measure from the floor sump of your bin to the roof and then to the outside and down to the location where you will put the HMT device. Then add (maybe) 20% to the length.
 - If you will be using the PRO version, you will need to do this for each sensor.
- Also, don't forget the pipe couplers and PVC glue



Stainless steel cable to hang the sensor end of the PVC pipe inside the bin.

- Something like "1/8 in. x 250 ft. Galvanized Aircraft Cable, 7x7 Construction - 340 lbs Safe Work Load - Reeled"



- Don't forget the cable clamps

You will also need the necessary hardware to hang the PVC either along the sidewall or at the top of the bin and along the outside of the bin going down to the HMT device's mounting location.

- Bolts/Nuts, Eye Bolts, Washers, etc.
- 4x4 Post Base
 - These worked well for us as we mounted our PVCs on the bin's sidewall



Tools You Will Need

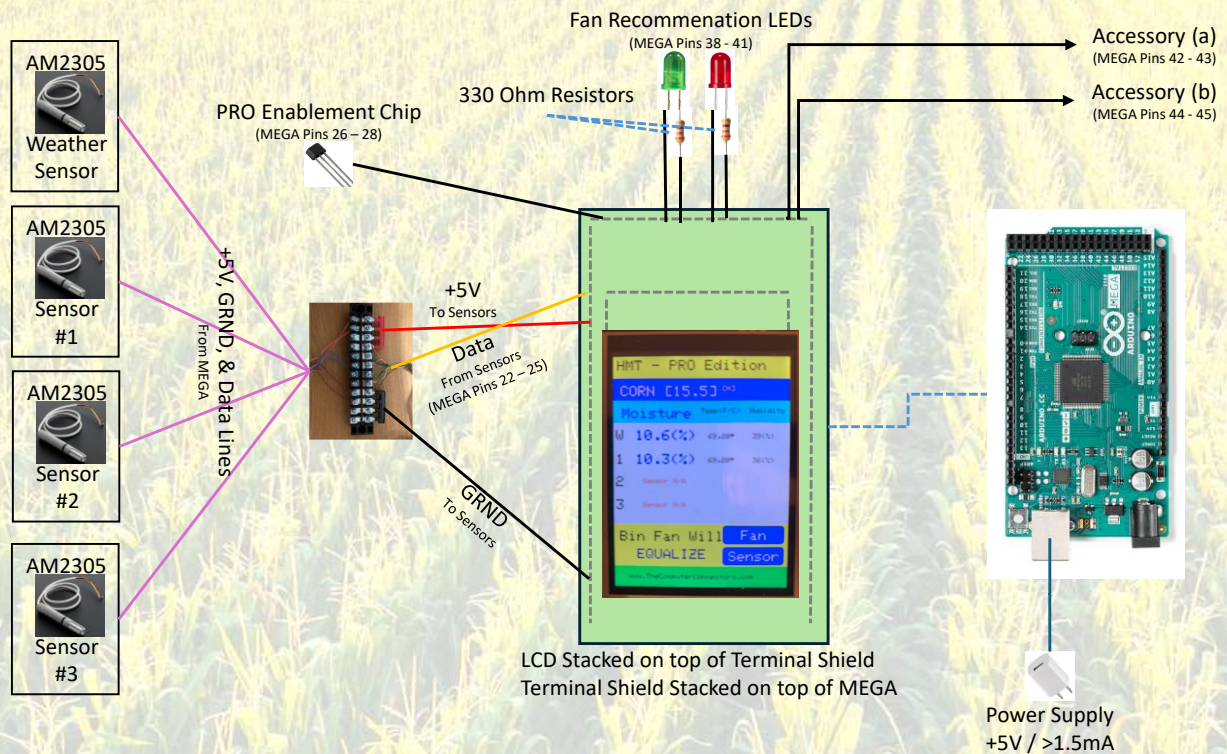
- Your reading glasses 😊
- A flashlight
- Ladder & Harness
 - It'd be a REALLY good idea to wear a harness while climbing on the bin.
- Soldering iron and solder



- Drill and various bits
- A variety of screws (for hanging the unit & the junction box)
- Needle nose pliers
- Wire stripper/cutter
- Knife
- Electrical tape
- Shrink tubing (optional)
- Hot glue gun (optional – to help secure the device's components and wiring)
- Caulk - (optional – to help secure and protect the device's components and wiring)
 - Also, don't forget to seal any holes you drill in the bin.
- Various sized wrenches
- Screw drivers
 - Including a really small, "jewelry type" flat blade
- Plastic pull-ties

Assembling the Device

Schematic



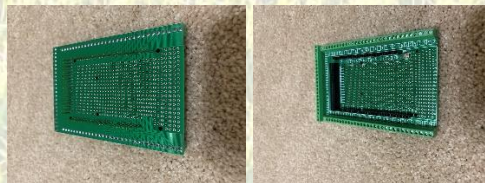
Assembling the Pieces and Placing Them in the Box

Start with the MEGA circuit board.



CAREFULLY, layer the terminal shield on top of the MEGA circuit board.

DO NOT FORCE these two items together! There are **LOTS** of pins that need to line up exactly before the entire terminal shield can be fully inserted onto the MEGA. If you use too much force, you may bend or even break the pins on the terminal shield. It took us, and will take you, a while to get things lined up perfectly. It can be done! Just be gentil and use lots of patience.



Layer the LCD on top of the terminal shield.

Regarding the LCDs orientation on top of the terminal shield, ensure that the SD card slot on the bottom of the LCD is located near the USB connection on the MEGA board.

TEXT DISPLAY NOTE: Recently, we received notification from a few builders that their TFT screens were incorrectly displaying all text and images. In these cases, everything was working except that all the text and images were rotated 180° from the touch locations. Thus, we added an “orientation jumper” feature to the software. Now, you can jumper either pins 33-34 or 34-35 to change the orientation so that the text and images match your touch locations.

Just like the terminal block’s mounting on top of the MEGA, DO NOT FORCE these two items together! Even though there are less pins that need to line up, you don’t want to bend or break any.

IMPORTANT NOTE: NEVER ever, NEVER ever press on the glass portion of the LCD. It will break! If you look closely at our picture, we made that mistake and cracked the screen. **Luckily, for us, it still works except that the touch features are finicky.** You may not be that lucky! Again, don’t make the mistake that we made! Just use a gentil hand and lots of patience.

Make Your Own Grain Bin Monitoring System



- VERY CAREFULLY, drill a hole at the bottom of your outer case.
 - Use a bit that is just large enough to fit the USB cable's rounded end.
- Insert the USB cable through the box hole and connect it to the Mega circuit board.
- Use the hot glue gun (or other method) to secure MEGA circuit board (in its case) to the outer case.
- Those three things layered together and inserted into the box result in this:



Sensor Placement Decisions

1. Determine the number of sensors you will install in your bin.
 - a. If you are using the FREE version, you will have just one sensor in the bin.
 - b. If you plan to purchase the PRO version, you will have up to three sensors inside the bin.
2. Determine where in the bin you would like your sensors.
 - a. Near bottom (center), near middle (slightly outward), and near top (closer to sidewall) was our choice.

Assembling and Mounting the Interior PVC Pipes

You may find this part of the device construction project the most time-consuming. It involves (for EACH sensor):

For each sensor inside the bin...

1. Determine the length of the PVC pipe you will need inside the bin. You may hang the pipe any way you deem best.

IMPORTANT NOTE: When determining your sensor placement, be mindful of the potential data cable length necessary to reach the HMT device. We tested our devices with data cable lengths up to about 100 feet. At that length, the sensor still worked for us, but we still would not suggest using a sensor cable with a length beyond 50 to 70 feet.

- a. When we installed our sensors, we measured a straight line from the “man hole” entrance at the top of the bin’s sidewall to the middle of the bin just a few feet above the floor sump.
 - i. **Our theory is:** When unloading the bin, the grain flows down from the top first but then quickly begins pulling at an angle from the side to the middle. If the PVC pipe is angled roughly in-line with the pull of the grain, we will have less of a chance of the grain flow forces damaging the installation.
 - ii. Thus, if we angle the PVC from the “man hole” to the middle of the bin, the grain’s horizontal flow forces during bin unloading will be minimized.
 1. The vertical flow forces will still be considerable.
 - iii. **DON'T UNDERESTIMATE** the force of the grain pulling on the PVC. You wouldn't believe it if we told you. Ensure that your PVC mounts are secure.
2. Now, determine how your sensor cable will exit the bin.
3. Determine the length of the PVC pipe necessary to protect the cable from the “man hole” to the exit point.
 - a. *In some cases, you may not even need this piece. If the cable will be following the inside wall of the bin to its exit point, you may not need the protective PVC. Just be careful to not connect the cable to anywhere that the grain can pull on it during bin unloading.*
4. Construct these PVC pipes using PVC couplers and PVC glue.
5. Drill holes all the way through one end of the pipe.
 - a. A bolt will be used to mount the PVC to the 4x4 Post Mount.



i.

6. Drill two sets of holes all the way through both ends of the pipe.
 - a. These holes will be used to secure the sensor at the end of the pipe and to attach the supporting cable.

Note the string in the following pictures. We thought that it would be a good idea to have the string available for when we need to replace the sensor and pull its data cable through the PVC.



7. For the PRO version, REPEAT the process for each sensor.

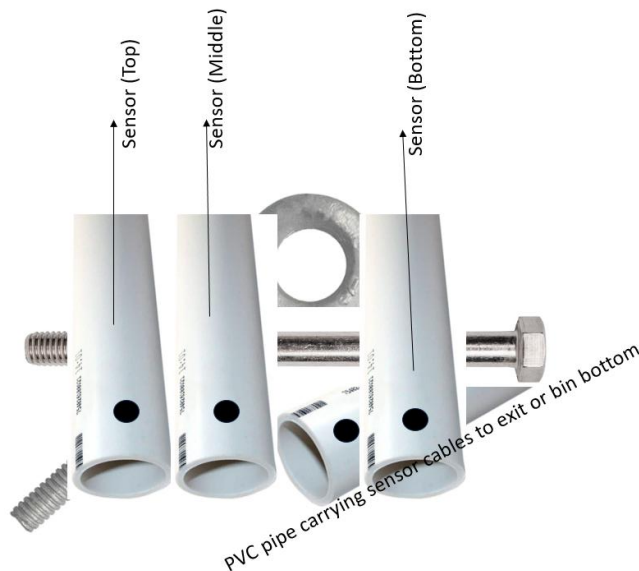


As an alternative mounting strategy, we attached an eyebolt to the bin's manhole frame and then used a bolt to secure the PVC pipes to the eyebolt. However, the "post mount" strategy shown above seems to be easier and more secure.

In our case [PRO version], we used three PVC pipes inside the bin. [In this mock-up] The three similarly oriented PVC pipes carry individual, low-voltage, sensor cables. The other pipe carries all three cables to the bin exit.

We drilled holes through BOTH ends of all of the PVC pipes. The "holes" you see in this mock-up, are used along with a long bolt to connect them together.

The long bolt is also inserted through an eye bolt which is used to attach the entire assembly to a reinforced part of the bin near the side man-hole.



(Un)Common – HMT

Make Your Own Grain Bin Monitoring System



Assembling the Exterior PVC Pipes

1. On the outside of the bin, determine the path and PVC pipe length necessary to protect the cable from the bin's exit point to the point where you will hang the HMT device.
 - a. **Keep in mind that you should hang the device out of direct sunlight.**
2. Add up all of those PVC pipe lengths.
 - a. The sum **(plus several extra feet)** will roughly represent the length of cable you will need for this sensor.
3. Construct the necessary PVC pipes using PVC couplers and PVC glue.

Connecting the Sensors to the Low-Voltage Cable & PVC Pipe Assembly

Preparing the Sensors

Here is something we just learned...

For years, we have been using AM2305 sensors directly out of the box, and we have never had a problem. Unfortunately, these original AM2305 sensors are nearly impossible to find. Even when you specifically order AM2305 sensors, you will probably receive AM2305B sensors.

The manufacturer says that the AM2305B sensor is a better and more reliable sensor, and we can't argue that point. However, this newer sensor version does require additional assembly work before it will work correctly. So, here is what you need to do...

1. Get your soldering iron, solder, and 4.7K ohm resistors
2. Heat up the iron
3. Solder the 4.7k ohm resistor between the sensor's V+ and data wires.
4. Do this for each sensor.

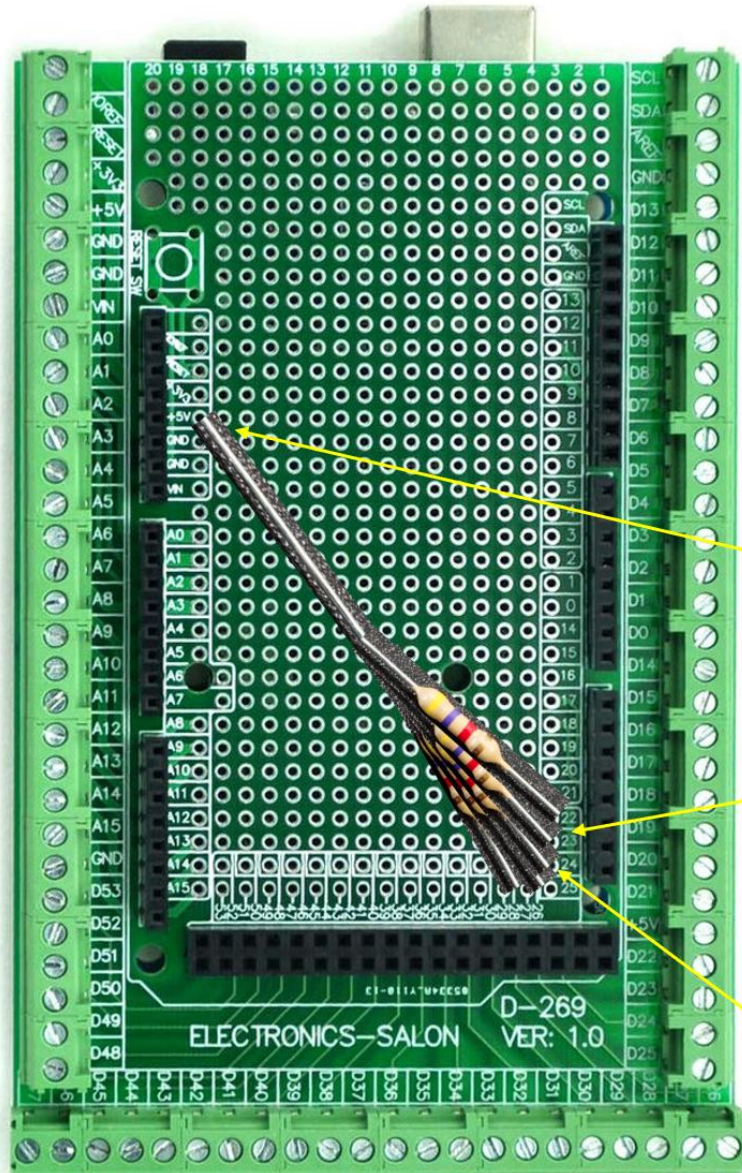


Perhaps you are wondering why this is necessary. Well, I can't explain why the old sensor version didn't require this. However, as a general explanation:

The sensor's data connection works by cycling the signal from V+ to GRND in a pattern that the Arduino MEGA can then interpret into temperature and humidity information. The sensor's internal circuitry isn't always able to pull the signal all the way up to full V+ voltage. So, the resistor is there to help pull the voltage up.

Preparing the Sensors (Alternative Method)

If you are **really good** with a soldering iron and you want a nice clean “look” to your device, you could solder the 4.7K ohm resistors directly to the Terminal Shield. See the following image.



- Carefully solder the other ends of the (5) 4.7K ohm resistors together and then to the +5V port on the Terminal Shield.

- Carefully solder one end of (4) 4.7K ohm resistors to ports 22 through 25 on the Terminal Shield.

- Be careful to appropriately insulate the metal leads on each resistor to ensure that they don't touch each other and short out their individual port connections.

Measuring, Cutting, and Installing the Cable

For each PVC pipe and sensor...

1. Using the PVC pipe and low-voltage cable measurements from above, cut the appropriate length of low-voltage cable to reach all the way through both the interior and exterior PVC pipes **PLUS** whatever you will need at the end of the exterior pipe to connect to the HMT device box.
 - a. In other words, you want a piece of data cable with a length **NO LESS THAN** what you measure from the sensor hanging in the bin, up to the roof, over to the man-hole, exiting the bin, down the side of the bin, and over to the HMT mounting point.
2. From the top end of the **internal** PVC pipe, thread the low-voltage cable down through the PVC pipe.
3. When the cable exits the bottom of the PVC pipe, solder (or use a wire splice connector) the sensor's +5V, Ground, and Data conductors to the low-voltage cable.
 - a. CAT5 (or similar) cable typically has 8 color coded conductors.
 - b. Technically, you may use any of those 8 colored conductors. Just be consistent – and **DOCUMENT what you did**.
 - c. Use electrical tape or shrink tubing to protect the soldered connections.
 - d. **Tip:** Your installation will look more professional if you are consistent with the colors you use. **Perhaps:**

Function	Sensor Cable Color	Data Cable Color
+5V	RED	ORANGE
Ground	WHITE	BROWN
SENSOR DATA	YELLOW	BLUE



4. At the end of the PVC pipe, secure the sensor into the pipe with a pull-tie.
 - a. Leave just about $\frac{1}{4}$ to $\frac{1}{2}$ " of the sensor sticking out of the pipe.
 - b. See the following image.

Sensor exits at end of each PVC pipe.
Leave about $\frac{1}{4}$ to $\frac{1}{2}$ " of the sensor sticking out of the PVC pipe.



Use a plastic tie through the drilled hole to secure the sensor in the PVC pipe.

5. Thread the other end of the low-voltage cable through the PVC pipes that run from the mounting point to the bin exit.
6. Measure the distance from the top of the bin to the point above the floor where the sensor will hang within the PVC pipe.
7. Using that measurement, cut a length of stainless-steel cable.
8. Carefully hang the end of the PVC from the top of the bin.
 - a. Ensure that the stainless steel cable does not interfere with anything else (ex. Your bin's fill Spreader)
 - b. The PVC should now be hanging more/less diagonally from its mounting point to the sensor's final location within the bin.
9. Label the data cables coming from each PVC with something that is at least semi-permanent to indicate the sensor's location within the bin. This is especially important with the PRO version since it has up to 3 interior sensors. That means three data cables. Next year, will you remember which cable is top, middle, or bottom? **Don't overlook this step! You'll thank us later.**
10. Thread the remaining data cable through the bin exit and to the HMT device location.

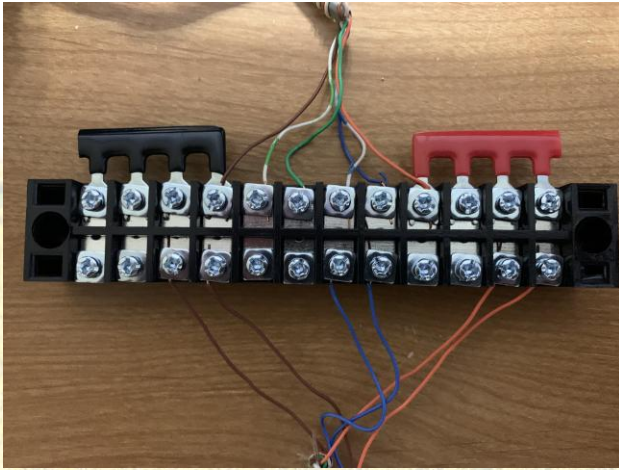
Connecting the HMT Device to the 12 Position Terminal Block

1. Cut a piece of data cable that is long enough to reach from the HMT device to the electrical junction box (plus a little extra).
2. Mount the terminal block inside of the junction box.

NOTE: We are using CAT3 cable for our installation. If you use a different type of cable, the wire colors will vary from below.

3. Strip several inches of the data cable's outer insulation from both ends of the cable.
4. Untwist the 8 revealed data cable conductors.
5. Connect the ORANGE wire to one of the +5V terminals on the HMT device's terminal shield.
6. Connect the BROWN wire to one of the Ground terminals on the HMT device's terminal shield.
7. Connect the BLUE wire to the HMT device's terminal shield D22 terminal.
8. Connect the BLUE/WHITE wire to the HMT device's terminal shield D23 terminal.
9. Connect the GREEN wire to the HMT device's terminal shield D24 terminal.
10. Connect the GREEN/WHITE wire to the HMT device's terminal shield D25 terminal.
11. Connect the ORANGE wire to FOUR of the terminal block's connectors. (See picture.)
 - a. Yup, FOUR
 - b. These four connectors on the terminal block will power the four sensors.
12. Connect the BROWN wire to FOUR of the terminal block's connectors. (See picture.)
 - a. Yup, FOUR
 - b. These four connectors on the terminal block will provide a Ground connection for the four sensors.
13. Connect the BLUE wire to one of the terminal block's connectors.
 - a. This connection will connect to the "Weather" sensor's data wire.
14. Connect the BLUE/WHITE wire to one of the terminal block's connectors.
 - a. This connection will connect to sensor 1's data wire.
15. Connect the GREEN wire to one of the terminal block's connectors.
 - a. This connection will connect to sensor 2's data wire.
16. Connect the GREEN/WHITE wire to one of the terminal block's connectors.
 - a. This connection will connect to sensor 3's data wire.

Make Your Own Grain Bin Monitoring System



- Notice how we have connected the HMT's +5V wire (orange) to four posts in the block and how we have connected the HMT's Ground wire to four posts in the block.
- The sensors' +5V and Ground wires get connected individually in the block.
- In this picture, the weather and sensor 1's data wires (blue) are connected to the blue and white/blue conductors in the block.
 - In this picture, the green conductors are unused and reserved for sensors 3 and 4.

Alternatively, if you ordered the Wire Splice Connectors, use those instead of the Terminal Block.



Connecting the Sensors to the 12 Position Terminal Block

Connecting the Weather Sensor to the Terminal Block

1. Cut a piece of data cable that is long enough to reach from the Weather sensor's mounting location to the junction box (plus a little extra).
2. Strip several inches of the data cable's outer insulation from both ends of the cable.
3. Untwist the 8 revealed data cable conductors.
4. Solder the RED wire from the Weather sensor to the data cable's ORANGE wire.
5. Solder the WHITE (or BLACK) wire from the Weather sensor to the data cable's BROWN wire.
6. Solder the YELLOW wire from the Weather sensor to the data cable's BLUE wire.

At the other end of the data cable:

7. Connect the ORANGE wire to one of the +5V connections on the terminal block.
8. Connect the BROWN wire to one of the Ground connections on the terminal block.
9. Connect the BLUE wire to the BLUE wire that you already connected to the terminal block.

Connecting Sensors 1 – 3 to the 12 Position Terminal Block

For each of the three sensors:

1. Strip several inches of the data cable's outer insulation from both ends of the cable.
2. Untwist the 8 revealed data cable conductors.
3. Solder the RED wire from the sensor to the data cable's ORANGE wire.
4. Solder the WHITE (or BLACK) wire from the sensor to the data cable's BROWN wire.
5. Solder each sensor's data connection to the terminal block.
 - a. BLUE wire from sensor 1 to the BLUE/WHITE connection on the terminal block.
 - b. BLUE wire from sensor 2 to the GREEN connection on the terminal block.
 - c. BLUE wire from sensor 3 to the GREEN/WHITE connection on the terminal block

Connecting LED and Accessory Cable to Terminal Block

Regardless of whether you plan to use the Accessory ports, it is a good idea to connect a cable to them now. If you don't add this cable now, you may need to do some device disassembly/reassembly later if you chose to use them.

1. Cut a piece of data cable that is long enough to reach from the HMT device to the junction box (plus a little extra).
2. Strip several inches of the data cable's outer insulation from both ends of the cable.
3. Untwist the 8 revealed data cable conductors.
4. Connect the ORANGE wire to pin 38 on the terminal shield.
5. Connect the ORANGE/WHITE wire to pin 39 on the terminal shield.
6. Connect the BROWN wire to pin 40 on the terminal shield.
7. Connect the BROWN/WHITE wire to pin 41 on the terminal shield.
8. Connect the BLUE wire to pin 42 on the terminal shield.

Make Your Own Grain Bin Monitoring System

9. Connect the BLUE/WHITE wire to pin 43 on the terminal shield.
10. Connect the GREEN wire to pin 44 on the terminal shield.
11. Connect the GREEN/WHITE wire to pin 45 on the terminal shield.
12. Route the other end of the data cable to the junction box.

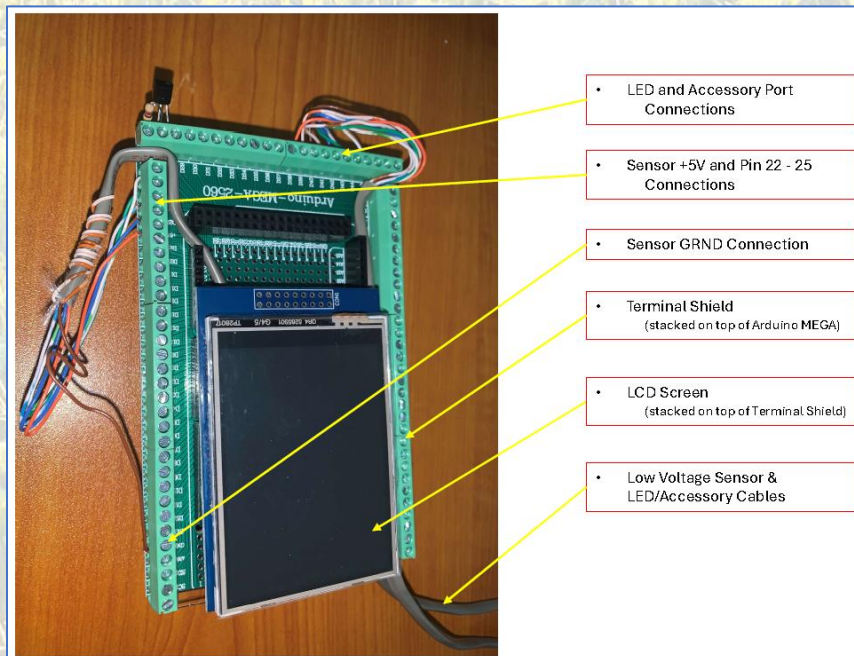
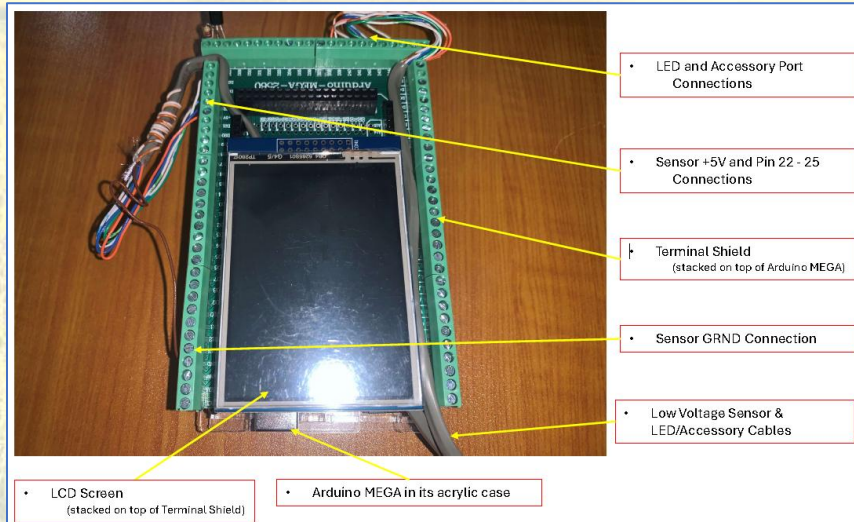
Alternatively, if you ordered the Wire Splice Connectors, use those instead of the Terminal Block.



Admiring What You Have Accomplished

Your device should look a little bit like our unfinished device image below.

Verify that you have all the wires connected securely and correctly.



A photograph of a custom-built grain bin monitoring system. The system is housed in a blue plastic enclosure. Inside, a Raspberry Pi is connected to a breadboard with various electronic components, including a sensor module. The Raspberry Pi screen displays a web interface with data. Below the enclosure, there are two grey cylindrical sensors connected by wires to a terminal block. The background is a blurred image of a cornfield under a blue sky.

Theoretically, your device should now be assembled to the point where it would work – except that we haven't loaded the firmware yet.

Page 36 of 47

©2026, The Computer Connectors LLC. All rights reserved.

www.TheComputerConnectors.com

August 27, 2026



A photograph of a custom-built grain bin monitoring system. The system is housed in a blue plastic enclosure. Inside, a green circuit board is visible, featuring a small LCD screen displaying text. Various wires are connected to the board and a terminal block. Two grey cylindrical sensors are connected to the system via cables. The background shows a vast field of golden corn under a clear blue sky.

Theoretically, your device should now be assembled to the point where it would work – except that we haven't loaded the firmware yet.

Page 36 of 47

©2026, The Computer Connectors LLC. All rights reserved.

www.TheComputerConnectors.com

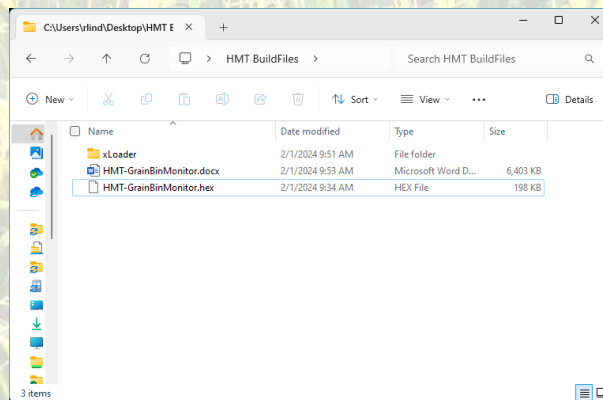
August 27, 2026

Installing the Firmware

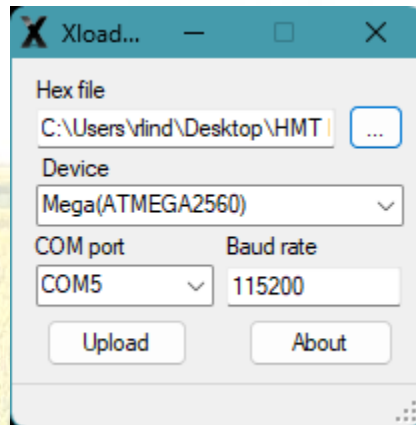
NOTE: If you are not comfortable using a computer, just take these instructions and your previously built HMT device to your local computer shop/geek. They should be able to get everything working for you.

The steps to installing the firmware are:

1. Build a FREE version by following the steps above.
2. Using a PC computer, download firmware from our website
 - a. www.TheComputerConnectors.com
3. The download will be in the form of a .ZIP which will include a couple files.
 - a. The software (Xloader) you will use to load the firmware on to the MEGA processor board
 - b. The latest firmware (.HEX file)
4. Using the USB cable, connect the HMT device to one of your computer's USB ports.
5. Create a folder on your computer to store the .ZIP file's contents.
 - a. Using your mouse, right-click any open (not occupied by an icon) spot on your computer's desktop.
 - b. Select NEW and Folder
 - c. Give the new folder a name (whatever you want)
 - i. We will call it "HMT BuildFiles"
6. Open the .ZIP file you downloaded above.
7. Copy the files from inside the .ZIP file to the new folder you created above.



8. Open the xLoader folder.
9. Double-click the "XLoader.exe" file to run the XLoader program.



a.

In the XLoader software:

10. Under the “Hex File” label, find the .HEX file you extracted from the downloaded .ZIP file and saved to your computer.
11. Under the “Device” label, select “MEGA (ATMEGA2560).”
12. Under the “COM port” label, select the COM port that your computer assigned to the HMT device.
 - a. It may be different than what you see in the image above.
13. Under the “Baud Rate” label, set the program to use “115200” baud.
14. Click “Upload”
 - a. The upload process should take about a minute.
15. When the upload is complete, you should see the message “xxxxxx bytes uploaded” at the bottom of the XLoader window.
16. **CONGRATULATIONS!**
 - a. **You have successfully loaded the firmware for your new HMT device.**



Installing the “Pro” Version License (optional)

Why are we selling the PRO version?

Honestly, we are just hoping to sell some PRO licenses to help offset the research and development costs that went into designing and distributing this device.

The steps to installing the PRO license are:

1. **Build a FREE version and verify that it is fully operational.**
 - a. You may upgrade to PRO at any time. Even after installation.
2. Order a PRO License from our website: www.TheComputerConnectors.com
 - a. If you wish to source the LEDs and resistors (see the build instructions above) from us, please include that option during your order checkout process.
 - b. Ensure that you provide us with a valid email and mailing address.
3. If you ordered the LED and resistor components from us during checkout, the installation instructions will be included with the PRO Version Enablement Components (below).
4. We will send you an email to verify your order, and we will mail you (most likely via USPS) the additional components and instructions necessary to upgrade to PRO.

Preliminary Testing

If all has gone well, the HMT device should be reading the sensors now.



If it is not, check all your wiring and refer to the “Troubleshooting” section at the end of this document.

Finalizing the Installation

1. If you haven't already, finished hanging the exterior PVC pipe from the bin exit to the HMT mounting location.
2. Mount the HMT device in its permanent location.
3. Mount the Electrical Box near the HMT device - Upside down.
4. Place and secure the weather sensor in the Electrical Box.
 - a. The electrical box's purpose is to protect the sensor while still allowing it full access to the outside air.
5. Use the hot glue gun, caulk, pull ties, or whatever is necessary to secure all the assembled components.
6. Visually inspect everything.
7. Fix anything that "doesn't look right."

Final Testing

1. Connect your new HMT device to a power source.
2. Is the device reading all the installed sensors and displaying Moisture, Temperature, and Humidity values for each sensor?
 - a. See the Troubleshooting section if something isn't working.
 - b. If yes, you are ready to build additional devices for your other bins.
 - c. If no, our strategy is to:
 - i. Take a coffee break.
 - ii. Verify all the wiring.
 - iii. Work through the Troubleshooting section.

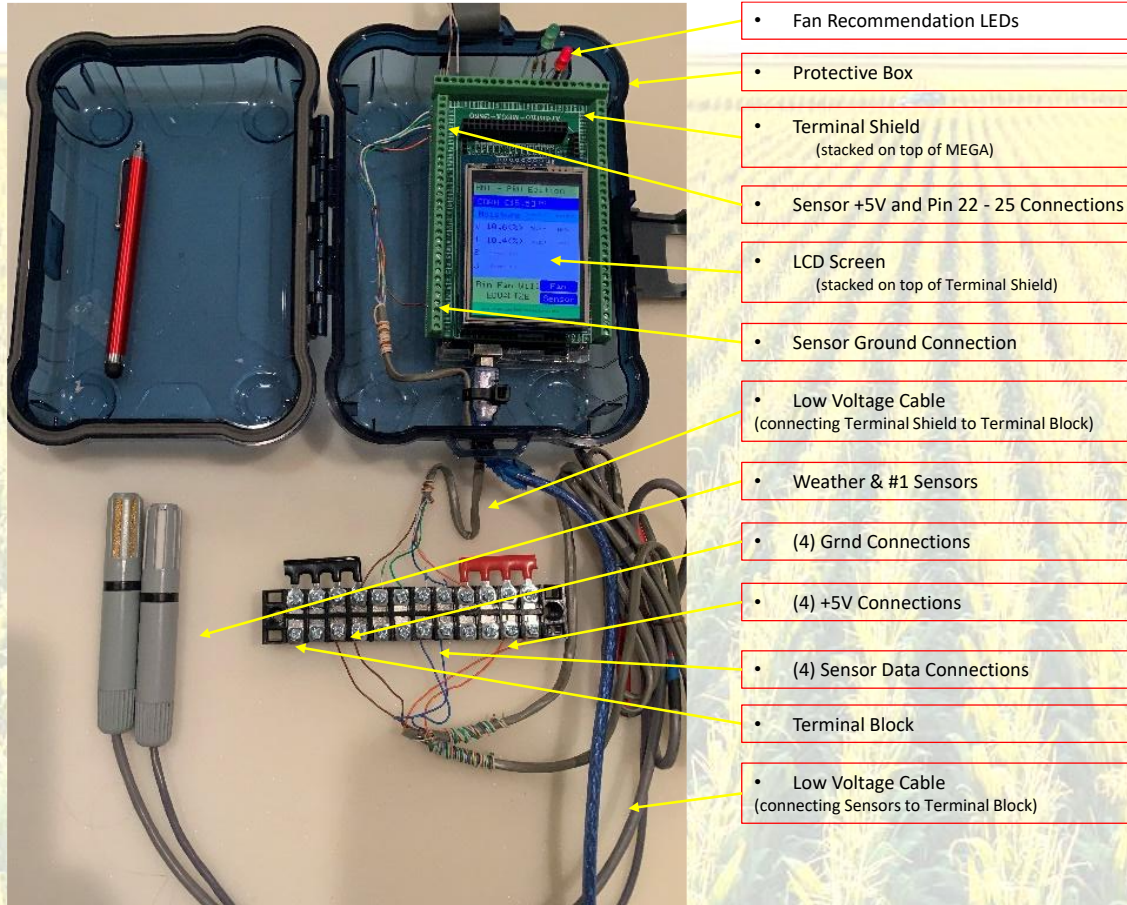
Congratulations on a job well done!

NOW – Go brag about your successful project to all your friends and neighbors!

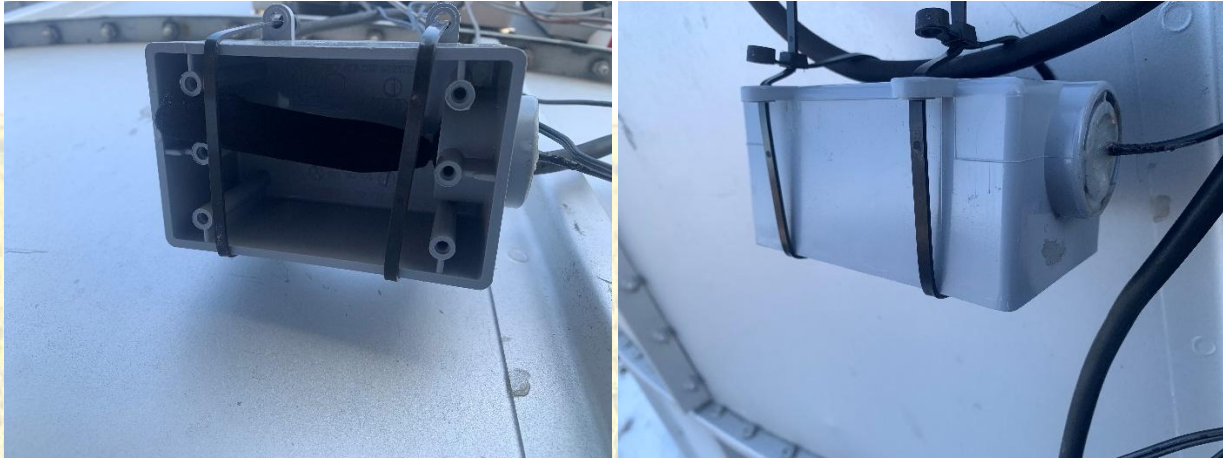
APPENDICES

Appendix A: Our Build and Installation Pictures

The HMT Device



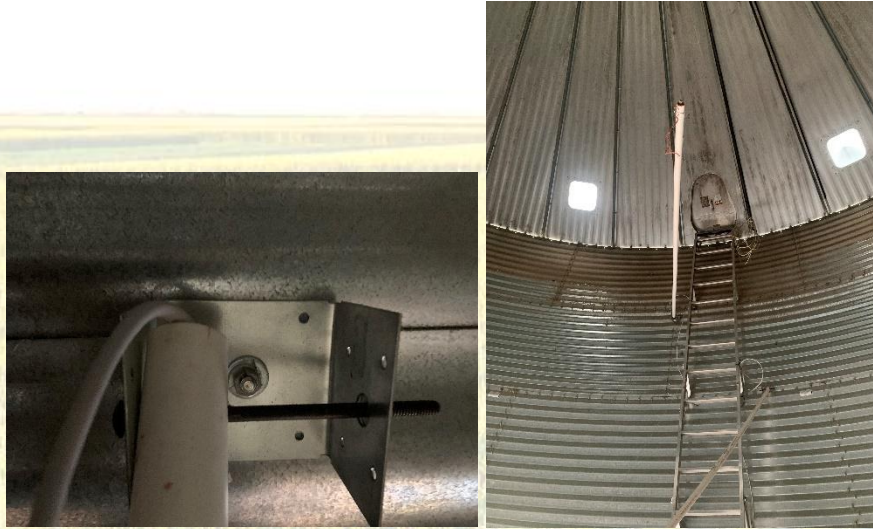
Weather Sensor – Protected by Electrical Box



In the above two pictures, we are showing you the Electrical Box that we are using to protect the AM2305 Weather Sensor. A few things to notice:

- The box is hung so that the opening is facing down. In this example, we simply used plastic pull ties to hang it from a cable.
- We placed the sensor in the box through the hole that, normally, an electrical conduit would connect.
- We then sealed the hole with silicone caulk.
- You may also notice that we added a sleeve over the sensor. The sleeve is just a piece of breathable fabric. We believe that this simple piece of fabric will help keep the sensor clean, and thus, extend its life.

Hanging the Sensor PVC Pipes



In the above two pictures, we are showing you how we mounted one of the interior PVC pipes that we are using to protect the AM2305 Sensor cables. In this case, we are using the 4x4 Post Mount as a mounting base for the PVC pipe.

Alternatively...



Sensor exits at end of each PVC pipe.
Leave about 1" of the sensor sticking out of the PVC pipe.



Use a plastic tie through the drilled hole to secure the sensor in the PVC pipe.

In the above two pictures, we are showing you the interior PVC pipes that we are using to protect the AM2305 Sensor cables. A few things to notice:

- In this bin, we are using two sensors. One located near the floor sump. The second located about 2/3 of the way up from the floor.
- The third PVC pipe is carrying (protecting) both sensor cables down to the bin exit. That bin exit is located near the bottom of the bin. Depending upon how you decide to exit the bin, you may or may not need a similar PVC pipe.

Make Your Own Grain Bin Monitoring System

- Notice how we are using a long bolt to secure all three pipes and an eye bolt together.
- In this bin, the eye bolt is securing the sensor pipes to the frame of the bin's side man-hole.
- Since the bin is about half full of soybeans, you can't see that we have plastic pull-tied the two sensor PVC pipes together at a couple places along their lengths. We believe that this simple act of connecting them together will help provide additional support.
- Since the bin is about half full of soybeans, you also cannot see that there is a stainless steel cable hanging from the bin's top man-hole down to the end of the bottom sensor's PVC pipe. We are using that cable to hang and better secure the end of the bottom sensor's PVC pipe near the floor sump.
- In the third [mockup] picture, we are using two holes and a plastic pull-tie at the end of the sensors' PVC pipes to secure the sensors in the pipes. Leave about an inch of the sensor sticking out of the pipe.
 - Just like we did with the Weather Sensor in the above picture, you may wish to cover the sensor with a piece of breathable fabric to help keep it clean and extend its life.
 - Remember!... Replace and Calibrate the sensors often!

Protecting Some of the Outside Equipment



Ok – don't laugh. We finished this installation about a week before harvest. So, we were in a bit of a rush. We intend to come back and "re-engineer" this outside portion of our installation sometime soon. Anyway...

We used a simple "battery box" to store several items:

- Surge protector
- Excess wire
- Power supply.

Knowing that a little moisture (snow or rain) could find its way into the box, we also drilled a hole in the bottom of the box.

A Word From Our Sponsor

A Word From Our Sponsor

Thank you for your interest in this project. Even though (or perhaps because) a lot of time and money went into the research and development, it is very rewarding to see it come to fruition.

We hope that you had some fun building this device. We certainly did!

Tell all your friends and neighbors!

Please let us know about any challenges or ideas you have. We are always looking for ways to improve. As such, we do plan on occasionally releasing new firmware updates. And, in a lot of cases, those firmware updates will add new features. And of course, the firmware updates are FREE. So, visit our website (www.TheComputerConnectors.com) often.

Our Other Products and Services

Also on our website, you will find links to our other products and services. You will find:

- Technology Services
- Business Consulting Services
- Training Services
- Drone Services (Certified FAA Unmanned Pilot)
- Software Products
- Technology Books such as:
 - How Computer Work
 - Microsoft Word
 - Microsoft Excel
 - Microsoft Windows
 - See them at: <http://amazon.com/author/ronaldind>.
- We even have a New Zealand travel book available.